

JEMMALI MOHAMED MAYSSEN

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Profile Summary

Final-Year Software Engineering Student at ESPRIT with hands-on experience in AI engineering, backend systems, and applied machine learning. Experienced in building data workflows, RAG pipelines, and practical, user-facing AI applications. Seeking a **6-month End-of-Studies Internship** in AI Engineering starting January 2027.

Professional Experience

BeeCoders Tunis, Tunisia
AI & Software Engineering Intern Jun 2026 - Aug 2026

- Designed and developed **SmartPFE**, an end-to-end AI platform guiding students through their graduation thesis lifecycle: topic ideation, report generation, and **Gemini-based jury simulation**.
- Implemented a **Corrective RAG pipeline** over **3,092 chunks from 31 PFE reports**, combining Python ingestion, multilingual embeddings, **MongoDB Atlas Vector Search**, retrieval grading, query rewriting, and resilient fallbacks.
- Ran a **RAGAS evaluation** across **10 bilingual scenarios**, revealing grounding gaps; the corrective pipeline rewrote queries in **5/10** cases. Traced LLM calls with **Langfuse**.
- Implemented a transactional **AI-credit system** with pre-request reservation, success-based settlement, automatic refunds, and shadow-mode rollout for production testing.

QantumShift Tunis, Tunisia
ML & Software Engineering Intern Jul 2025 - Aug 2025

- Developed a three-tier marketplace using **Angular**, **Spring Boot REST APIs**, **MongoDB**, and **JWT authentication**.
- Built a recommendation engine from user interactions using **NMF**, **cosine similarity**, and a fallback for **cold-start users**.

TUNISAIR Tunis, Tunisia
Full-Stack Developer Intern (Bachelor's Graduation Project) Jan 2024 - Jun 2024

- Delivered a centralized **B2B portal** for travel agencies, covering contract management, claims, financial reporting, and credit workflows.
- Integrated **Amadeus flight data** and implemented **JWT authentication**, **RBAC authorization**, and audit logging using **Symfony**, **PHP**, and **MySQL**.

Selected Project

DeepSkyn - AI Skincare Analysis Platform Jan 2026 - May 2026
1st Place, ESPRIT Project Fair

- Built an award-winning platform combining **Gemini multimodal photo analysis**, personalized report and routine generation, classical machine learning, and computer vision.
- Benchmarked **five supervised-learning models** on **10,000 records**; selected **Random Forest** at approximately **81% accuracy**, **0.73 macro-F1**, and **0.91 macro-AUC**.
- Trained a **MobileNetV2 transfer-learning CNN** on **14,839 skin images**, achieving **90.4% validation accuracy** across four skin-type classes.

Education

ESPRIT - Private Higher School of Engineering and Technology Tunis, Tunisia
Engineering Degree in Software Engineering 2024 - Expected 2027
Faculty of Sciences of Tunis Tunis, Tunisia
Bachelor of Science in Computer Science 2021 - 2024

Technical Skills

AI & ML: Python, Gemini API, RAG/CRAG, RAGAS, Langfuse, embeddings, vector search, Sentence Transformers, scikit-learn, TensorFlow, CNNs, recommender systems.
Backend & Data: Node.js, Express, TypeScript, Java, Spring Boot, MongoDB, PostgreSQL, MySQL, REST APIs.
Engineering: Docker, Git, CI/CD, JWT, OAuth, React, Angular.

Languages

Arabic: Native
English: Professional Proficiency
French: Working Proficiency